

StockNet - Parallelized Neural Network for Stock Price Prediction Using High-Performance Computing Resources

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Abstract

In this project, we aim to design and implement a stock price predictor using the HPC cluster. Our approach will focus on the S&P 500 historical dataset, which provides a good source of financial data for model training and evaluation. We will begin by collecting and preprocessing this data to make sure it is clean, consistent, and suitable for modeling. Using the computational power of the Markov cluster, we will develop and train multiple neural network models to predict stock price movements and identify stocks likely to outperform market trends. These models will be designed to capture nonlinear relationships and temporal dependencies within the market.

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1 Introduction

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